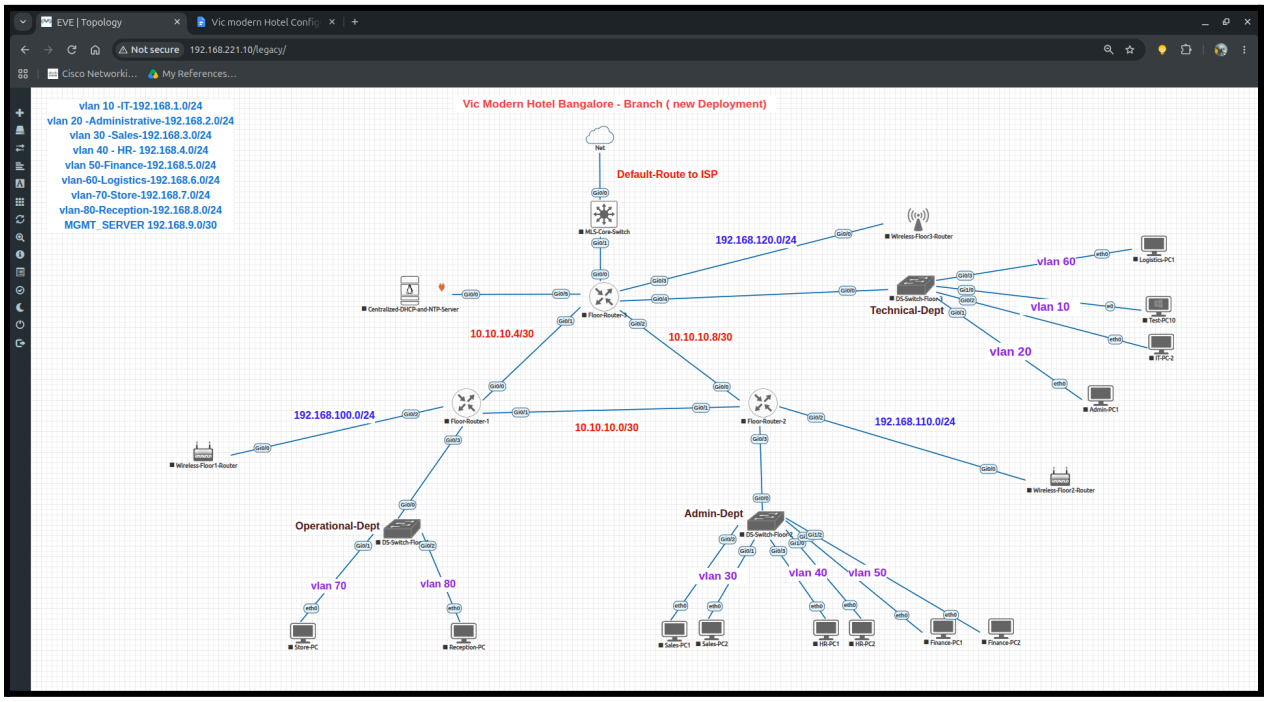
Vic Modern Hotel Network Deployment Configurations

Eve-NG Topology:



Vlan Configuration Commands:

Floor1- Distribution Switch:

- en
- Config t
- hostname DS-Switch-Floor1
- banner motd .
  Hi This Switch is the Floor 1 Operation Department Acting as a Distribution Switch and Access Switch

.

- vlan 70
- name store
- exit

- vlan 80
- name Reception
- Exit

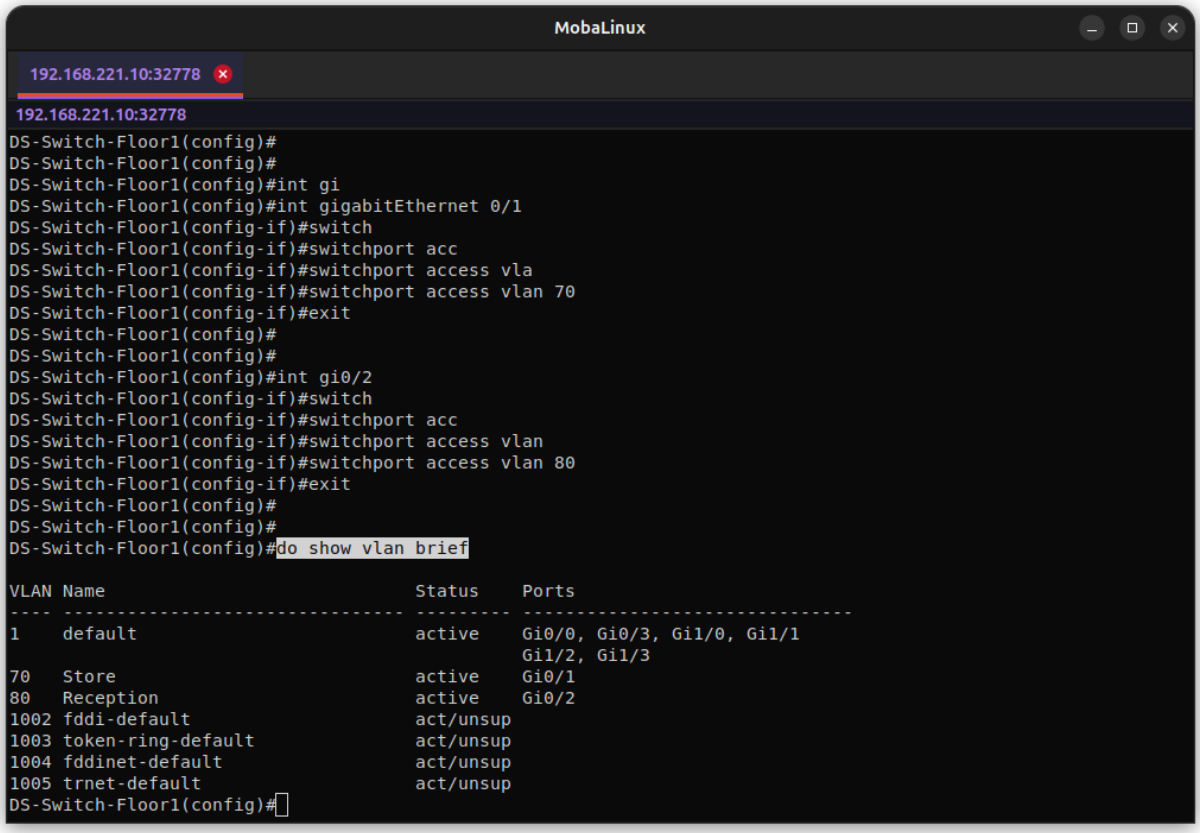
Assigning AccessPorts to Respective Vlan:

DS-Switch-Floor1(config)#int gigabitEthernet 0/1
DS-Switch-Floor1(config)#:switchport access vlan 70
DS-Switch-Floor1(config)#:exit

DS-Switch-Floor1(config)#int gi0/2
DS-Switch-Floor1(config-if)#switchport access vlan 80
DS-Switch-Floor1(config-if)#exit

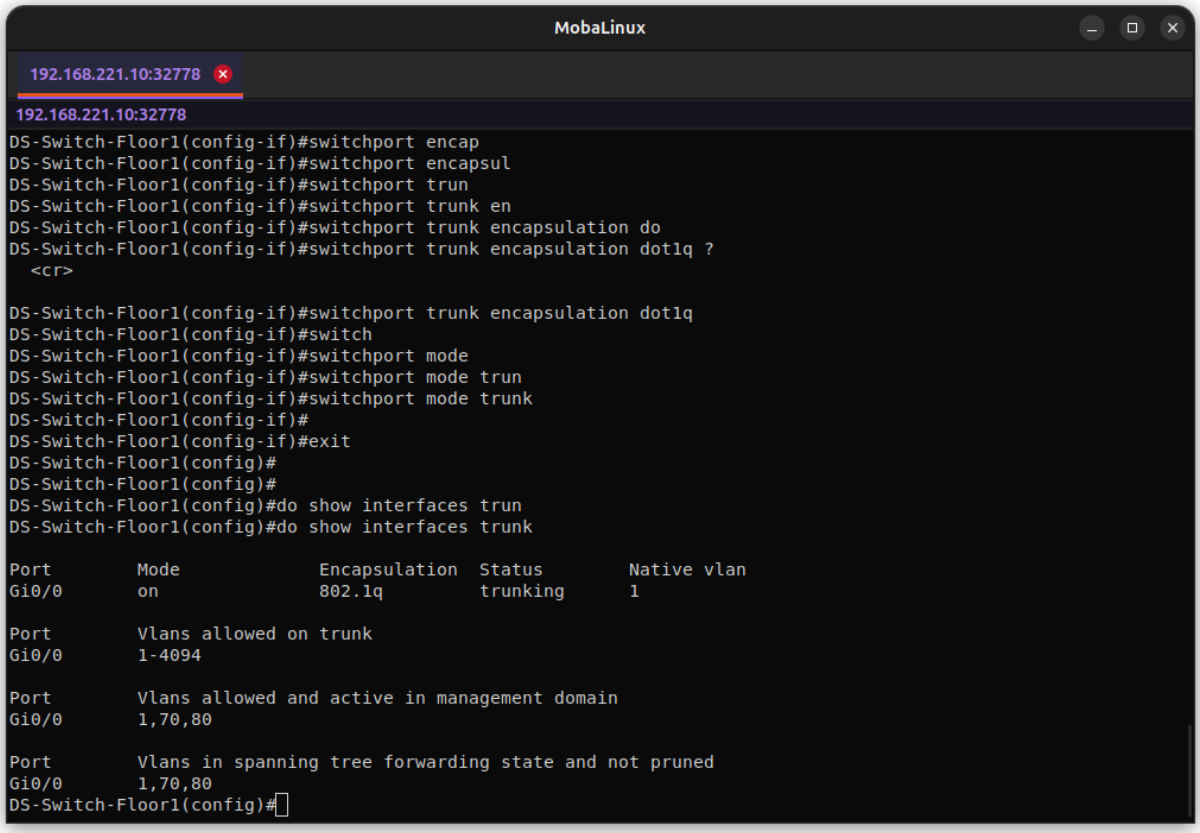
Verification Commands:

do show vlan brief



Configuring trunk Interfaces Towards Router:

DS-Switch-Floor1(config)#int gi0/0
DS-Switch-Floor1(config-if)#switchport trunk encapsulation dot1q
DS-Switch-Floor1(config-if)#switchport mode trunk
DS-Switch-Floor1(config-if)#exit



Configuration for Floor2-Switch:

- en
- Config t
- hostname DS-Switch-Floor1
- banner motd .
  Hi This Switch is Floor2 Switch Configured as the DS and Access Switch for the Floor 2
  .
  DS-Switch-Floor2(config)#vlan 30
  DS-Switch-Floor2(config-vlan)#name Sales
  DS-Switch-Floor2(config-vlan)#exit

```
DS-Switch-Floor2(config)#vlan 40
DS-Switch-Floor2(config-vlan)#name HR
DS-Switch-Floor2(config-vlan)#exit

DS-Switch-Floor2(config)#vlan 50
DS-Switch-Floor2(config-vlan)#name Finance
DS-Switch-Floor2(config-vlan)#exit
```

Assigning AccessPorts to Respective Vlan:

```
DS-Switch-Floor2(config)#int range gigabitEthernet 0/1-2
DS-Switch-Floor2(config-if-range)#switchport access vlan 30
DS-Switch-Floor2(config-if-range)#exit

DS-Switch-Floor2(config)#int range gigabitEthernet 0/3, gigabitEthernet 1/0
DS-Switch-Floor2(config-if-range)#switchport access vlan 40
DS-Switch-Floor2(config-if-range)#
DS-Switch-Floor2(config-if-range)#exit

DS-Switch-Floor2(config)#int range gigabitEthernet 1/1-2
DS-Switch-Floor2(config-if-range)#switchport access vlan 50
DS-Switch-Floor2(config-if-range)#exit
```

Trunk Configuration on the Switches Floor 2

```
DS-Switch-Floor2(config)#int gi0/0
DS-Switch-Floor2(config-if)#Switchport trunk encapsulation dot1q
DS-Switch-Floor2(config-if)#switchport mode trunk
DS-Switch-Floor2(config-if)#exit
```

Switch 3 floor 3 Configurations:

```
DS-Switch-Floor3(config)#vlan 10
DS-Switch-Floor3(config-vlan)#name IT
DS-Switch-Floor3(config-vlan)#exit

DS-Switch-Floor3(config)#vlan 20
DS-Switch-Floor3(config-vlan)#name Administrative
DS-Switch-Floor3(config-vlan)#exit


DS-Switch-Floor3(config)#vlan 60
DS-Switch-Floor3(config-vlan)#name Logistics
DS-Switch-Floor3(config-vlan)#exit
```

Assigning Access Ports to the Respective Vlan:

```
DS-Switch-Floor3(config)#int gi0/1
DS-Switch-Floor3(config-if)#switchport access vlan 20
DS-Switch-Floor3(config-if)#exit
DS-Switch-Floor3(config)#

DS-Switch-Floor3(config)#int range gigabitEthernet 1/0, gigabitEthernet 0/2
DS-Switch-Floor3(config-if-range)#switchport access vlan 10
DS-Switch-Floor3(config-if-range)#exit

DS-Switch-Floor3(config)#int gigabitEthernet 0/3
DS-Switch-Floor3(config-if)#switchport access vlan 60
DS-Switch-Floor3(config-if)#
DS-Switch-Floor3(config-if)#exit
```

Trunk Configuration on the Switches Floor 3

```
DS-Switch-Floor2(config)#int gi0/0
DS-Switch-Floor2(config-if)#Switchport trunk encapsulation dot1q
DS-Switch-Floor2(config-if)#switchport mode trunk
DS-Switch-Floor2(config-if)#exit
```

Router Configurations:

Intervlan Routing for Floor 1 Main Networks:

```
en
conf t
ho Floor-RTR-1
Floor-RTR-1(config)#banner motd .
Enter TEXT message.  End with the character '.'.
Hi this Router is the Floor1-Router Inter & OSPF has Configured .

Floor-RTR-1(config)#
Floor-RTR-1(config)#int gigabitEthernet 0/3.1
Floor-RTR-1(config-subif)#encapsulation dot1Q 70
Floor-RTR-1(config-subif)#ip address
Floor-RTR-1(config-subif)#ip address 192.168.7.1 255.255.255.0
Floor-RTR-1(config-subif)#exit
Floor-RTR-1(config)#int gig
Floor-RTR-1(config)#int gigabitEthernet 0/3.2
Floor-RTR-1(config-subif)#encapsulation dot1Q 80
Floor-RTR-1(config-subif)#ip address 192.168.8.1 255.255.255.0
Floor-RTR-1(config-subif)#exit
Floor-RTR-1(config)#
```

Floor 2:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#ho Floor-RTR-2
Floor-RTR-2(config)#
Floor-RTR-2(config)#banner mot
Floor-RTR-2(config)#banner motd .
Hi this Router is Acting as a Floor 2 for OSPF routing and Inter-vlan ROuting
.
Floor-RTR-2(config)#int gigabitEthernet 0/3.1
Floor-RTR-2(config-subif)#encapsulation dot1Q 30
Floor-RTR-2(config-subif)#ip address 192.168.3.1 255.255.255.0
Floor-RTR-2(config-subif)#exit

Floor-RTR-2(config)#int gigabitEthernet 0/3.2
Floor-RTR-2(config-subif)#encapsulation dot1Q 40
Floor-RTR-2(config-subif)#ip address 192.168.4.1 255.255.255.0
Floor-RTR-2(config-subif)#exit

Floor-RTR-2(config)#int gigabitEthernet 0/3.3
Floor-RTR-2(config-subif)#encapsulation dot1Q 50
Floor-RTR-2(config-subif)#ip address 192.168.5.1 255.255.255.0
Floor-RTR-2(config-subif)#exit
```

Floor 3:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#ho Floor-RTR-3
Floor-RTR-3(config)#bann
Floor-RTR-3(config)#banner mo
Floor-RTR-3(config)#banner motd .
Enter TEXT message.  End with the character '.'.
Hi this Router is for Floor 3 Acting as a OSPF routing and Inter-vlan ROuting
.
Floor-RTR-3(config)#int gi0/4
Floor-RTR-3(config-if)#no shut
Floor-RTR-3(config-if)#exit

Floor-RTR-3(config)#int gi0/4.1
Floor-RTR-3(config-subif)#encapsulation dot1Q 10
Floor-RTR-3(config-subif)#ip address 192.168.1.1 255.255.255.0
Floor-RTR-3(config-subif)#exit

Floor-RTR-3(config)#int gigabitEthernet 0/4.2
Floor-RTR-3(config-subif)#encapsulation dot1Q 20
Floor-RTR-3(config-subif)#ip address 192.168.2.1 255.255.255.0
Floor-RTR-3(config-subif)#exit

Floor-RTR-3(config)#int gigabitEthernet0/4.3
Floor-RTR-3(config-subif)#encapsulation dot1Q 60
Floor-RTR-3(config-subif)#ip address 192.168.6.1 255.255.255.0
Floor-RTR-3(config-subif)#exit
```

Configuring DHCP Server for Main Networks:

```
Router(config)#ho DHCP-NTP-SERVER
DHCP-NTP-SERVER(config)#

DHCP-NTP-SERVER(config)#int gi0/0
DHCP-NTP-SERVER(config-if)#ip address 192.168.9.2 255.255.255.252
DHCP-NTP-SERVER(config-if)#no shut
DHCP-NTP-SERVER(config-if)#exit
DHCP-NTP-SERVER(config)#

DHCP-NTP-SERVER(config)#
DHCP-NTP-SERVER(config)#ip dhcp pool FLOOR1-VLAN70
DHCP-NTP-SERVER(dhcp-config)#network 192.168.7.0 255.255.255.0
DHCP-NTP-SERVER(dhcp-config)#default-router 192.168.7.1
DHCP-NTP-SERVER(dhcp-config)#dns-server 8.8.8.8
DHCP-NTP-SERVER(dhcp-config)#domain-name vic.com
DHCP-NTP-SERVER(dhcp-config)#lease 1 23
DHCP-NTP-SERVER(config)#ip dhcp excluded-address 192.168.7.1


DHCP-NTP-SERVER(config)#ip dhcp pool FLOOR1-VLAN80
DHCP-NTP-SERVER(dhcp-config)#network 192.168.8.0 255.255.255.0
DHCP-NTP-SERVER(dhcp-config)#default-router 192.168.8.1
DHCP-NTP-SERVER(dhcp-config)#dns-server 8.8.8.8
DHCP-NTP-SERVER(dhcp-config)#domain-name vic.com
DHCP-NTP-SERVER(config)#ip dhcp excluded-address 192.168.8.1
DHCP-NTP-SERVER(config)
```

```
DHCP-NTP-SERVER(config)#ip dhcp pool FLOOR2-VLAN30
DHCP-NTP-SERVER(dhcp-config)#network 192.168.3.0 255.255.255.0
DHCP-NTP-SERVER(dhcp-config)#default-router 192.168.3.1
DHCP-NTP-SERVER(dhcp-config)#dns-server 8.8.8.8
DHCP-NTP-SERVER(dhcp-config)#domain-name vic.com
DHCP-NTP-SERVER(dhcp-config)#lease 1
DHCP-NTP-SERVER(config)#ip dhcp excluded-address 192.168.3.1

DHCP-NTP-SERVER(config)#ip dhcp pool FLOOR2-VLAN40
DHCP-NTP-SERVER(dhcp-config)#network 192.168.4.0 255.255.255.0
DHCP-NTP-SERVER(dhcp-config)#default-router 192.168.4.1
DHCP-NTP-SERVER(dhcp-config)#dns-server 8.8.8.8
DHCP-NTP-SERVER(dhcp-config)#lease 1
DHCP-NTP-SERVER(dhcp-config)#exit
DHCP-NTP-SERVER(config)#ip dhcp excluded-address 192.168.4.1

DHCP-NTP-SERVER(config)#ip dhcp pool FLOOR2-VLAN50
DHCP-NTP-SERVER(dhcp-config)#network 192.168.5.0 255.255.255.0
DHCP-NTP-SERVER(dhcp-config)#default-router 192.168.5.1
DHCP-NTP-SERVER(dhcp-config)#dns-server 8.8.8.8
DHCP-NTP-SERVER(dhcp-config)#domain-name vic.com
DHCP-NTP-SERVER(dhcp-config)#lease 1
DHCP-NTP-SERVER(dhcp-config)#exit
DHCP-NTP-SERVER(config)#ip dhcp excluded-address 192.168.5.1

DHCP-NTP-SERVER(config)#ip dhcp pool FLOOR3-VLAN10
DHCP-NTP-SERVER(dhcp-config)#network 192.168.1.0 255.255.255.0
DHCP-NTP-SERVER(dhcp-config)#default-router 192.168.1.1
DHCP-NTP-SERVER(dhcp-config)#dns-server 8.8.8.8
DHCP-NTP-SERVER(dhcp-config)#domain-name vic.com
DHCP-NTP-SERVER(dhcp-config)#lease 1
DHCP-NTP-SERVER(dhcp-config)#exit
DHCP-NTP-SERVER(config)#ip dhcp excluded-address 192.168.1.1
```

DHCP-NTP-SERVER(config)#ip dhcp pool FLOOR3-VLAN20
DHCP-NTP-SERVER(dhcp-config)#network 192.168.2.0 255.255.255.0
DHCP-NTP-SERVER(dhcp-config)#default-router 192.168.2.1
DHCP-NTP-SERVER(dhcp-config)#dns-server 8.8.8.8
DHCP-NTP-SERVER(dhcp-config)#domain-name vic.com
DHCP-NTP-SERVER(dhcp-config)#lease 1
DHCP-NTP-SERVER(dhcp-config)#exit
DHCP-NTP-SERVER(config)#ip dhcp excluded-address 192.168.2.1

DHCP-NTP-SERVER(config)#ip dhcp pool FLOOR3-VLAN60
DHCP-NTP-SERVER(dhcp-config)#network 192.168.6.0 255.255.255.0
DHCP-NTP-SERVER(dhcp-config)#default-router 192.168.6.1
DHCP-NTP-SERVER(dhcp-config)#dns-server 8.8.8.8
DHCP-NTP-SERVER(dhcp-config)#domain-name vic.com
DHCP-NTP-SERVER(dhcp-config)#lease 1
DHCP-NTP-SERVER(dhcp-config)#exit
DHCP-NTP-SERVER(config)#ip dhcp excluded-address 192.168.6.1

Wireless gateway Router:

Wireless-Floor1-RTR(config)#int gi0/0
Wireless-Floor1-RTR(config-if)#ip add 192.168.100.2 255.255.255.0
Wireless-Floor1-RTR(config-if)#no shut
Wireless-Floor1-RTR(config-if)#exit

Wireless-Floor1-RTR(config)#ip dhcp pool WIRELESS-GUEST-FLOOR1
Wireless-Floor1-RTR(dhcp-config)#network 192.168.100.0 255.255.255.0
Wireless-Floor1-RTR(dhcp-config)#default-router 192.168.100.1
Wireless-Floor1-RTR(dhcp-config)#dns-server 8.8.8.8
Wireless-Floor1-RTR(dhcp-config)#domain-name vic.com
Wireless-Floor1-RTR(config)#ip dhcp excluded-address 192.168.100.1
192.168.100.2

```
WIRELESS-FLOOR2-RTR(config)#
WIRELESS-FLOOR2-RTR(config)#int gi0/0
WIRELESS-FLOOR2-RTR(config-if)#ip address 192.168.110.2 255.255.255.0
WIRELESS-FLOOR2-RTR(config-if)#no shut
WIRELESS-FLOOR2-RTR(config-if)#exit

WIRELESS-FLOOR2-RTR(config)#ip dhcp pool WIRELESS-FLOOR2-RTR
WIRELESS-FLOOR2-RTR(dhcp-config)#network 192.168.110.0 255.255.255.0
WIRELESS-FLOOR2-RTR(dhcp-config)#default-router 192.168.110.1
WIRELESS-FLOOR2-RTR(dhcp-config)#dns-server 8.8.8.8
WIRELESS-FLOOR2-RTR(dhcp-config)#lease 1
WIRELESS-FLOOR2-RTR(dhcp-config)#dom
WIRELESS-FLOOR2-RTR(dhcp-config)#domain-name vic.com
WIRELESS-FLOOR2-RTR(dhcp-config)#exit
WIRELESS-FLOOR2-RTR(config)#ip dhcp excluded-address 192.168.110.1
192.168.110.2
WIRELESS-FLOOR2-RTR(config)#


WIRELESS-FLOOR3_ROUT(config)#ho WIRELESS-FLOOR3_RTR
WIRELESS-FLOOR3_RTR(config)#int gi0/0
WIRELESS-FLOOR3_RTR(config-if)#ip add 192.168.120.2 255.255.255.0
WIRELESS-FLOOR3_RTR(config-if)#no shut
WIRELESS-FLOOR3_RTR(config-if)#exit


WIRELESS-FLOOR3_RTR(config)#ip dhcp pool WIRELESS-FLOOR3-RTR
WIRELESS-FLOOR3_RTR(dhcp-config)#network 192.168.120.0 255.255.255.0
WIRELESS-FLOOR3_RTR(dhcp-config)#default-router 192.168.120.1
WIRELESS-FLOOR3_RTR(dhcp-config)#dns-server 8.8.8.8
WIRELESS-FLOOR3_RTR(dhcp-config)#do
WIRELESS-FLOOR3_RTR(dhcp-config)#domain-name vic.com
WIRELESS-FLOOR3_RTR(dhcp-config)#
WIRELESS-FLOOR3_RTR(dhcp-config)#exit
```

WIRELESS-FLOOR3_RTR(config)#ip dhcp excluded-address 192.168.120.1 192.168.120.2

Interface Configuration for All Main Routers:

Floor-RTR-1(config)#int gi0/2
Floor-RTR-1(config-if)#ip add 192.168.100.1 255.255.255.0
Floor-RTR-1(config-if)#no shut
Floor-RTR-1(config-if)#exit

Floor-RTR-1(config)#int gi0/1
Floor-RTR-1(config-if)#ip add 10.10.10.1 255.255.255.252
Floor-RTR-1(config-if)#no shut
Floor-RTR-1(config-if)#exit
Floor-RTR-1(config)#

Floor-RTR-1(config)#int gi0/0
Floor-RTR-1(config-if)#ip add 10.10.10.5 255.255.255.252
Floor-RTR-1(config-if)#no shut
Floor-RTR-1(config-if)#exit
Floor-RTR-1(config)#

Floor-RTR-2(config)#
Floor-RTR-2(config)#int gi0/2
Floor-RTR-2(config-if)#ip add 192.168.110.1 255.255.255.0
Floor-RTR-2(config-if)#no shut
Floor-RTR-2(config-if)#exit

loor-RTR-2(config)#int gi0/1
Floor-RTR-2(config-if)#ip add 10.10.10.2 255.255.255.252
Floor-RTR-2(config-if)#no shut
Floor-RTR-2(config-if)#exit

Floor-RTR-2(config)#int gi0/0

```
Floor-RTR-2(config-if)#ip add 10.10.10.10 255.255.255.252
Floor-RTR-2(config-if)#no shut
Floor-RTR-2(config-if)#exit

Floor-RTR-3(config)#int gi0/1
Floor-RTR-3(config-if)#ip add 10.10.10.6 255.255.255.252
Floor-RTR-3(config-if)#no shut
Floor-RTR-3(config-if)#exit

Floor-RTR-3(config)#int gi0/2
Floor-RTR-3(config-if)#ip add 10.10.10.9 255.255.255.252
Floor-RTR-3(config-if)#no shut

Floor-RTR-3(config-if)#int gi0/3
Floor-RTR-3(config-if)#ip add 192.168.120.1 255.255.255.0
Floor-RTR-3(config-if)#no shut
Floor-RTR-3(config-if)#exit
Floor-RTR-3(config)#

Floor-RTR-3(config)#int gi0/5
Floor-RTR-3(config-if)#ip add 192.168.9.1 255.255.255.252
Floor-RTR-3(config-if)#no shut
Floor-RTR-3(config-if)#exit
Floor-RTR-3(config)#

Floor-RTR-3(config)#int gi0/0
Floor-RTR-3(config-if)#ip address 192.168.9.5 255.255.255.252
Floor-RTR-3(config-if)#no shut
Floor-RTR-3(config-if)#
Floor-RTR-3(config-if)#
Floor-RTR-3(config-if)#exit
```

MLS Core Config:

```
Switch>en
Switch#conf t
Switch(config)#ho MLS-Core-Switch
MLS-Core-Switch(config)#int gi0/1
MLS-Core-Switch(config-if)#no switchport
MLS-Core-Switch(config-if)#ip address 192.168.9.6 255.255.255.252
MLS-Core-Switch(config-if)#no shut
MLS-Core-Switch(config-if)#ip routing


MLS-Core-Switch(config)#int gi0/0
MLS-Core-Switch(config-if)#no switchport
MLS-Core-Switch(config-if)#ip address 172.16.88.100 255.255.255.0
MLS-Core-Switch(config-if)#no shut
MLS-Core-Switch(config-if)#exit
```

Router OSPF Configurations:

Wireless Routers OSPF Configurations:

```
Floor-RTR-3(config)#router ospf 1
Floor-RTR-3(config-router)#network 192.168.9.4 0.0.0.3 area 0
Floor-RTR-3(config-router)#network 192.168.120.0 0.0.0.255 area 0
Floor-RTR-3(config-router)#network 192.168.9.0 0.0.0.3 area 0
Floor-RTR-3(config-router)#network 192.168.6.0 0.0.0.255 area 0
Floor-RTR-3(config-router)#network 192.168.1.0 0.0.0.255 area 0
Floor-RTR-3(config-router)#network 192.168.2.0 0.0.0.255 area 0
Floor-RTR-3(config-router)#network 10.10.10.4 0.0.0.3 area 0
Floor-RTR-3(config-router)#network 10.10.10.8 0.0.0.3 area 0
Floor-RTR-3(config-router)#exit
Floor-RTR-3(config)#
```

Floor 1 RTR OSPF Configurations:

Floor-RTR-1(config)#router ospf 1
Floor-RTR-1(config-router)#network 192.168.100.0 0.0.0.255 area 0
Floor-RTR-1(config-router)#network 192.168.7.0 0.0.0.255 area 0
Floor-RTR-1(config-router)#network 192.168.8.0 0.0.0.255 area 0
Floor-RTR-1(config-router)#network 10.10.10.0 0.0.0.3 area 0
Floor-RTR-1(config-router)#network 10.10.10.4 0.0.0.3 area 0
Floor-RTR-1(config-router)#exit

Floor 2 RTR OSPF Configurations:

Floor-RTR-2(config)#router ospf 1
Floor-RTR-2(config-router)#network 192.168.3.0 0.0.0.255 area 0
Floor-RTR-2(config-router)#network 192.168.4.0 0.0.0.255 area 0
Floor-RTR-2(config-router)#network 192.168.5.0 0.0.0.255 area 0
Floor-RTR-2(config-router)#network 192.168.110.0 0.0.0.255 area 0
Floor-RTR-2(config-router)#network 10.10.10.8 0.0.0.3 area 0
Floor-RTR-2(config-router)#network 10.10.10.8 0.0.0.3 area 0
Floor-RTR-2(config-router)#network 10.10.10.0 0.0.0.3 area 0
Floor-RTR-2(config-router)#

Wireless Routers OSPF Configurations:

Wireless-Floor1-RTR(config)#router ospf 1
Wireless-Floor1-RTR(config-router)#network 192.168.100.0 0.0.0.255 area 0

WIRELESS-FLOOR2-RTR(config)#router ospf 1
WIRELESS-FLOOR2-RTR(config-router)#network 192.168.110.0 0.0.0.255 area 0
WIRELESS-FLOOR2-RTR(config-router)#exit

WIRELESS-FLOOR3_RTR(config)#router ospf 1

WIRELESS-FLOOR3_RTR(config-router)#network 192.168.120.0 0.0.0.255 area 0
WIRELESS-FLOOR3_RTR(config-router)#

MLS Default Route Configuration:

Default information that means last resort okay: on multilayer switch:

router ospf 1
Network 192.168.4.0 0.0.0.3 area 0
Default information originate always
Exit

We are configuring a default route in OSPF to solve a fundamental networking problem: routers cannot forward data to destinations they do not explicitly know about, causing them to immediately drop traffic destined for the internet or external networks. By implementing the `default-information originate command` on your Multilayer Switch (MLS), we transform the switch into the "Gateway of Last Resort" for the entire floor [INDEX]. This single configuration tells the MLS to dynamically advertise a `0.0.0.0/0` path via OSPF to Router 1, Router 2, and Router 3 simultaneously [INDEX]. This automated injection completely eliminates the tedious, error-prone need to manually type static routes on every individual router in your topology. When a PC initiates a ping to an unknown external destination, the local router checks its OSPF-learned table, identifies the default route placeholder, and seamlessly forwards the packet up to the MLS. The MLS then matches this traffic against its own static connection and hands it off directly to the Internet Service Provider (ISP) [INDEX]. Ultimately, this design ensures uninterrupted external connectivity for your end-users while keeping your internal routing dynamic, scalable, and highly adaptive to future network changes.

Ip helper-address for the Main networks:

DHCP-NTP-SERVER(config)#$ip route 0.0.0 0.0.0.0 gigabitEthernet 0/0 192.168.9.1

Floor-RTR-1(config)#int gigabitEthernet 0/3.1
Floor-RTR-1(config-subif)#ip helper-address 192.168.9.2
Floor-RTR-1(config-subif)#exit

Floor-RTR-1(config)#int gigabitEthernet 0/3.2
Floor-RTR-1(config-subif)#ip helper-address 192.168.9.2
Floor-RTR-1(config-subif)#exit
Floor-RTR-1(config)#

Floor-RTR-2(config)#int gigabitEthernet 0/3.1
Floor-RTR-2(config-subif)#ip helper-address 192.168.9.2
Floor-RTR-2(config-subif)#exit

Floor-RTR-2(config)#int gigabitEthernet 0/3.2
Floor-RTR-2(config-subif)#ip helper-address 192.168.9.2
Floor-RTR-2(config-subif)#exit

Floor-RTR-2(config)#int gigabitEthernet 0/3.3
Floor-RTR-2(config-subif)#ip helper-address 192.168.9.2
Floor-RTR-2(config-subif)#exit

Floor-RTR-2(config)#

Floor-RTR-3(config)#int gigabitEthernet 0/4.1
Floor-RTR-3(config-subif)#ip helper-address 192.168.9.2
Floor-RTR-3(config-subif)#exit

Floor-RTR-3(config)#int gigabitEthernet 0/4.2
Floor-RTR-3(config-subif)#ip helper-address 192.168.9.2
Floor-RTR-3(config-subif)#exit

Floor-RTR-3(config)#int gigabitEthernet 0/4.3
Floor-RTR-3(config-subif)#ip helper-address 192.168.9.2
Floor-RTR-3(config-subif)#exit

Port Security for test PC IT team only:

DS-Switch-Floor3(config)#int range gigabitEthernet 0/2, gigabitEthernet 1/0
DS-Switch-Floor3(config-if-range)#switchport mode access
DS-Switch-Floor3(config-if-range)#switchport port
DS-Switch-Floor3(config-if-range)#switchport port-security
DS-Switch-Floor3(config-if-range)#switchport-security mac-address sticky
DS-Switch-Floor3(config-if-range)#$port-security violation shutdown
DS-Switch-Floor3(config-if-range)#exit

PAT Configuration on MLS:

```
MLS-Core-Switch(config)#
MLS-Core-Switch(config)#int gi0/1
MLS-Core-Switch(config-if)#ip nat inside
MLS-Core-Switch(config-if)#exit


MLS-Core-Switch(config)#int gi0/0
MLS-Core-Switch(config-if)#ip nat outside
MLS-Core-Switch(config-if)#exit
MLS-Core-Switch(config)#
```

PAT Access-list:

MLS-Core-Switch(config)#ip access-list extended 100
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.100.0 0.0.0.255 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.7.0 0.0.0.255 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.8.0 0.0.0.255 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.1.0 0.0.0.255 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.2.0 0.0.0.255 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.3.0 0.0.0.255 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.4.0 0.0.0.255 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.5.0 0.0.0.255 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.6.0 0.0.0.255 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.110.0 0.0.0.255 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.120.0 0.0.0.255 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.9.0 0.0.0.3 any
MLS-Core-Switch(config-ext-nacl)#permit ip 192.168.9.4 0.0.0.3 any
MLS-Core-Switch(config-ext-nacl)#permit ip 10.10.10.0 0.0.0.3 any
MLS-Core-Switch(config-ext-nacl)#permit ip 10.10.10.4 0.0.0.3 any
MLS-Core-Switch(config-ext-nacl)#permit ip 10.10.10.8 0.0.0.3 any
MLS-Core-Switch(config-ext-nacl)#

Ip nat inside source list 100 interface gigabitethernet 0/0 overload

The MLS NAT configuration is correct (ACL `permit any`, `ip nat inside` on Gi0/1, `ip nat outside` on Gi0/0, overload rule present), but the packets from your PCs leave the MLS with their private source address (`192.168.1.2`) instead of being translated to `172.16.88.20`. This was shown by the capture on `pnet1`, by the empty `show ip nat translations`, and by the silent NAT debug. The most likely reason is that the L3 switch image in EVE-NG forwards transit packets through the CEF fast path without sending them to the NAT process, whereas a router image handles NAT properly. I haven't proven this, since NAT worked for the MLS's own pings and the statistics showed some CEF-translated packets, so it remains a likely cause rather than a confirmed one. Because the packets arrive at the EVE-NG host untranslated, the host's reply has no route back

to the private network and goes out its default route instead, so the request is sent but no reply returns. The fix is to do the NAT on a router image placed between the MLS and the Net cloud, or to add return routes on the EVE-NG host.

SSH Configuration on All Routers and then MLS:

SSH Configuration on the Wireless Router Floor 1:

Wireless-Floor1-RTR(config)#ip domain-name vic.com
Wireless-Floor1-RTR(config)#enable secret vic@2005
Wireless-Floor1-RTR(config)#crypto key generate rsa modulus 1024
Wireless-Floor1-RTR(config)#ip ssh version 2
Wireless-Floor1-RTR(config)#line vty 0
Wireless-Floor1-RTR(config-line)#login local
Wireless-Floor1-RTR(config-line)#transport input ssh
Wireless-Floor1-RTR(config-line)#exit
Wireless-Floor1-RTR(config)#
Wireless-Floor1-RTR(config)#username vicmanager privilege 15 password vic@2005

SSH Configuration on the Wireless Router Floor 1

Floor-RTR-1(config)#ip domain-name vic.com
Floor-RTR-1(config)#enable secret vic@2005
Floor-RTR-1(config)#username vicmanager privilege 15 password vic@2005
Floor-RTR-1(config)#crypto key generate rsa modulus
Floor-RTR-1(config)#crypto key generate rsa modulus 1024
Floor-RTR-1(config)#ip ssh version 2
Floor-RTR-1(config)#line vty 0
Floor-RTR-1(config-line)#login local
Floor-RTR-1(config-line)#transport input ssh
Floor-RTR-1(config-line)#transport input ssh
Floor-RTR-1(config-line)#exit

SSH Configuration on the Wireless Router Floor 1:

Floor-RTR-2(config)#ip domain-name vic.com
Floor-RTR-2(config)#username vicmanager privilege 15 password vic@2005
Floor-RTR-2(config)#enable secret vic@2005
Floor-RTR-2(config)#crypto key generate rsa modulus 1024
Floor-RTR-2(config)#ip ssh version 2
Floor-RTR-2(config)#line vty 0
Floor-RTR-2(config-line)#login local
Floor-RTR-2(config-line)#transport input ssh

SSH Configuration on the Wireless Router Floor 2 :

WIRELESS-FLOOR2-RTR(config)#ip domain-name vic.com
WIRELESS-FLOOR2-RTR(config)#enable secret vic@2005
WIRELESS-FLOOR2-RTR(config)#$cmanager privilege 15 password vic@2005
WIRELESS-FLOOR2-RTR(config)#crypto key generate rsa modulus 1024
WIRELESS-FLOOR2-RTR(config)#ip ssh version 2
WIRELESS-FLOOR2-RTR(config)#line vty 0
WIRELESS-FLOOR2-RTR(config-line)#transport input ssh
WIRELESS-FLOOR2-RTR(config-line)#login local
WIRELESS-FLOOR2-RTR(config-line)#exit

SSH Configuration on the Router Floor 3:

Floor-RTR-3(config)#ip domain-name vic.com
Floor-RTR-3(config)#enable secret vic@2005
Floor-RTR-3(config)#username vicmanager privilege 15 password vic@2005
Floor-RTR-3(config)#crypto key generate rsa modulus 1024
Floor-RTR-3(config)#ip ssh version 2
Floor-RTR-3(config)#line vty 0
Floor-RTR-3(config-line)#transport input ssh
Floor-RTR-3(config-line)#login local
Floor-RTR-3(config-line)#exit
Floor-RTR-3(config)#

Floor-RTR-3(config)#

SSH Configuration on the Wireless Router Floor 3:

WIRELESS-FLOOR3_RTR(config)#ip domain-name vic.com
WIRELESS-FLOOR3_RTR(config)#$cmanager privilege 15 password vic@2005
WIRELESS-FLOOR3_RTR(config)#crypto key generate rsa modulus 1024
WIRELESS-FLOOR3_RTR(config)#line vty 0
WIRELESS-FLOOR3_RTR(config-line)#transport input ssh
WIRELESS-FLOOR3_RTR(config-line)#login local
WIRELESS-FLOOR3_RTR(config-line)#exit
WIRELESS-FLOOR3_RTR(config)#

SSH Configuration on the MLS Core Switch:

MLS-Core-Switch(config)#ip domain-name vic.com
MLS-Core-Switch(config)#crypto key generate rsa modulus 1024
MLS-Core-Switch(config)#username vicmanager privilege 15 password vic@2005
MLS-Core-Switch(config)#ip ssh version 2
MLS-Core-Switch(config)#line vty 0
MLS-Core-Switch(config-line)#transport input ssh
MLS-Core-Switch(config-line)#login local
MLS-Core-Switch(config-line)#exit
MLS-Core-Switch(config)#enable secret vic@2005

SSH Configuration on the DHCP NTP SERVER :

DHCP-NTP-SERVER(config)#ip domain-name vic.com
DHCP-NTP-SERVER(config)#ip ssh version 2
DHCP-NTP-SERVER(config)#enable secret vic@2005
DHCP-NTP-SERVER(config)#username vicmanager privilege 15 password vic@2005
DHCP-NTP-SERVER(config)#crypto key generate rsa modulus 1024
DHCP-NTP-SERVER(config)#ip ssh version 2

```
DHCP-NTP-SERVER(config)#LINE VTY 0
DHCP-NTP-SERVER(config-line)#transport input ssh
DHCP-NTP-SERVER(config-line)#login local
DHCP-NTP-SERVER(config-line)#exit
DHCP-NTP-SERVER(config)#
```

SSH Configuration on the Wireless Router Floor 3:

```
WIRELESS-FLOOR3_RTR(config)#ip domain-name vic.com
WIRELESS-FLOOR3_RTR(config)#enable secret vic@2005
WIRELESS-FLOOR3_RTR(config)#$cmanager privilege 15 password vic@2005
WIRELESS-FLOOR3_RTR(config)#crypto key generate rsa modulus 1024
WIRELESS-FLOOR3_RTR(config)#ip ssh version 2
WIRELESS-FLOOR3_RTR(config)#
WIRELESS-FLOOR3_RTR(config)#line vty 0
WIRELESS-FLOOR3_RTR(config-line)#transport input ssh
WIRELESS-FLOOR3_RTR(config-line)#login local
WIRELESS-FLOOR3_RTR(config-line)#EXIT
```

Floor 1 ACL:

```
Floor-RTR-1(config)#do show access-lists
Extended IP access list 100
    5 deny tcp any any eq 22 (1 match)
    10 permit ip 192.168.100.0 0.0.0.255 192.168.1.0 0.0.0.255
    20 permit ip 192.168.100.0 0.0.0.255 192.168.7.0 0.0.0.255
    30 permit ip 192.168.100.0 0.0.0.255 192.168.8.0 0.0.0.255
    40 deny ip 192.168.100.0 0.0.0.255 192.168.2.0 0.0.0.255
```

50 deny ip 192.168.100.0 0.0.0.255 192.168.3.0 0.0.0.255
60 deny ip 192.168.100.0 0.0.0.255 192.168.4.0 0.0.0.255
70 deny ip 192.168.100.0 0.0.0.255 192.168.5.0 0.0.0.255
80 deny ip 192.168.100.0 0.0.0.255 192.168.6.0 0.0.0.255
90 deny ip 192.168.100.0 0.0.0.255 192.168.7.0 0.0.0.255
100 deny ip 192.168.100.0 0.0.0.255 192.168.9.0 0.0.0.3
120 permit ip any any (72 matches)

SSH BLOCK FOR NON IT VLAN ALSO:

Floor-RTR-1(config)#ip access-list extended SSH-BLK-NONIT
Floor-RTR-1(config-ext-nacl)#deny tcp any any eq 22
Floor-RTR-1(config-ext-nacl)#permit ip any any
Floor-RTR-1(config-ext-nacl)#exit
Floor-RTR-1(config)#int gi0/3
Floor-RTR-1(config-if)#ip access-group SSH_BLK-NONIT in
Floor-RTR-1(config-if)#exit

Floor 2 ACL:

Floor-RTR-2(config)#ip access-list extended 101
Floor-RTR-2(config-ext-nacl)#deny t
Floor-RTR-2(config-ext-nacl)#deny tcp
Floor-RTR-2(config-ext-nacl)#deny tcp any any eq 22
Floor-RTR-2(config-ext-nacl)#$.168.110.0 0.0.0.255 192.168.2.0 0.0.0.255
Floor-RTR-2(config-ext-nacl)#$.168.110.0 0.0.0.255 192.168.3.0 0.0.0.255
Floor-RTR-2(config-ext-nacl)#$.168.110.0 0.0.0.255 192.168.4.0 0.0.0.255
Floor-RTR-2(config-ext-nacl)#$.168.110.0 0.0.0.255 192.168.5.0 0.0.0.255
Floor-RTR-2(config-ext-nacl)#$.168.110.0 0.0.0.255 192.168.6.0 0.0.0.255
Floor-RTR-2(config-ext-nacl)#permit ip any any
Floor-RTR-2(config-ext-nacl)#exit

Floor 2 access-lists:

```
Floor-RTR-2(config)#do show access-lists
Extended IP access list 101
    10 deny tcp any any eq 22
    20 deny ip 192.168.110.0 0.0.0.255 192.168.2.0 0.0.0.255
    30 deny ip 192.168.110.0 0.0.0.255 192.168.3.0 0.0.0.255
    40 deny ip 192.168.110.0 0.0.0.255 192.168.4.0 0.0.0.255
    50 deny ip 192.168.110.0 0.0.0.255 192.168.5.0 0.0.0.255
    60 deny ip 192.168.110.0 0.0.0.255 192.168.6.0 0.0.0.255
    70 permit ip any any
Floor-RTR-2(config)#int gi0/2
Floor-RTR-2(config-if)#ip access-group 101 in
Floor-RTR-2(config-if)#
Floor-RTR-2(config-if)#exit
```

SSH-BLK FOR NONIT:

```
Floor-RTR-2(config)#ip access-list extended SSH-BLK_NONIT
Floor-RTR-2(config-ext-nacl)#deny tcp any any eq 22
Floor-RTR-2(config-ext-nacl)#permit ip any any
Floor-RTR-2(config-ext-nacl)#
Floor-RTR-2(config-ext-nacl)#exit


Floor-RTR-2(config)#
Floor-RTR-2(config)#int gi0/3
Floor-RTR-2(config-if)#ip access-group SSH-BLK_NONIT in
Floor-RTR-2(config-if)#exit
```

Floor 3 :

```
Floor-RTR-3(config)#ip access
Floor-RTR-3(config)#ip access-list ext
Floor-RTR-3(config)#ip access-list extended 102
```

```
Floor-RTR-3(config-ext-nacl)#

Floor-RTR-3(config-ext-nacl)#deny tcp any any eq 22
Floor-RTR-3(config-ext-nacl)#$.168.120.0 0.0.0.255 192.168.2.0 0.0.0.255
Floor-RTR-3(config-ext-nacl)#$.168.120.0 0.0.0.255 192.168.3.0 0.0.0.255
Floor-RTR-3(config-ext-nacl)#$.168.120.0 0.0.0.255 192.168.4.0 0.0.0.255
Floor-RTR-3(config-ext-nacl)#$.168.120.0 0.0.0.255 192.168.5.0 0.0.0.255
Floor-RTR-3(config-ext-nacl)#$.168.120.0 0.0.0.255 192.168.6.0 0.0.0.255
Floor-RTR-3(config-ext-nacl)#$.168.120.0 0.0.0.255 192.168.9.0 0.0.0.3
Floor-RTR-3(config-ext-nacl)#permit ip any any
Floor-RTR-3(config-ext-nacl)#exit
```

Assigning the ACL on the interface:

```
Floor-RTR-3(config)#int gi0/3
Floor-RTR-3(config-if)#ip access
Floor-RTR-3(config-if)#ip access-group 102 in
Floor-RTR-3(config-if)#
Floor-RTR-3(config-if)#exit
```

Access-List:

```
Floor-RTR-3(config)#do show access-lists
Extended IP access list 102
    10 deny tcp any any eq 22
    20 deny ip 192.168.120.0 0.0.0.255 192.168.2.0 0.0.0.255
    30 deny ip 192.168.120.0 0.0.0.255 192.168.3.0 0.0.0.255
    40 deny ip 192.168.120.0 0.0.0.255 192.168.4.0 0.0.0.255
    50 deny ip 192.168.120.0 0.0.0.255 192.168.5.0 0.0.0.255
    60 deny ip 192.168.120.0 0.0.0.255 192.168.6.0 0.0.0.255
    70 deny ip 192.168.120.0 0.0.0.255 192.168.9.0 0.0.0.3
    80 permit ip any any (1 match)
```

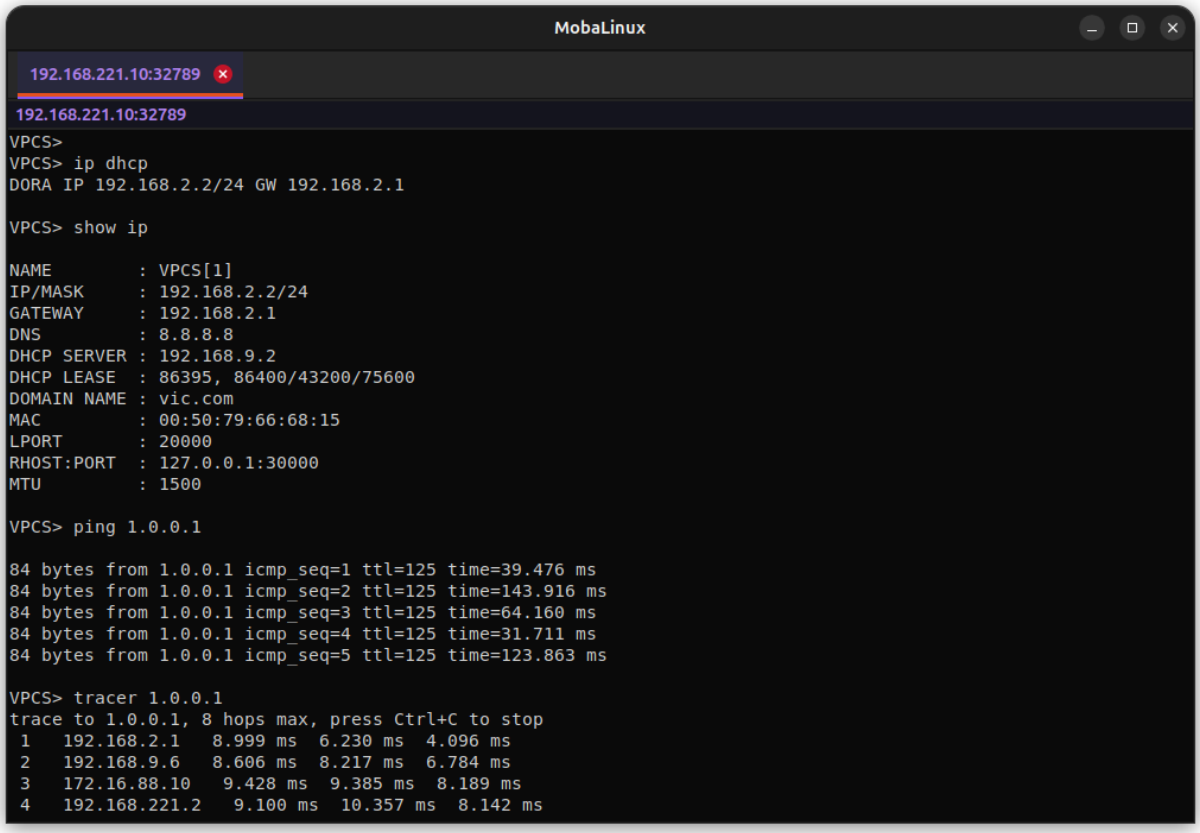
Floor 3 SSH-BLOCK-NONIT:

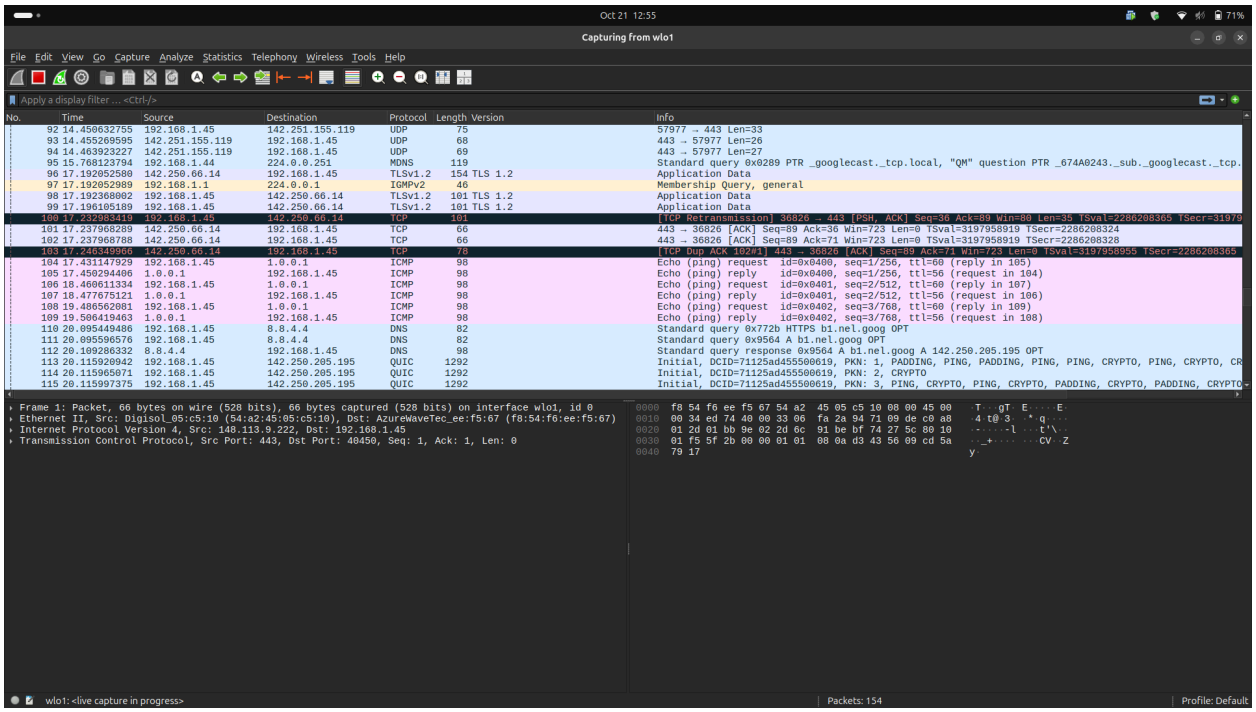
```
Floor-RTR-3(config)#ip access-list extended SSH-BLOCK-NONIT
Floor-RTR-3(config-ext-nacl)#deny tcp 192.168.6.0 0.0.0.255 any
Floor-RTR-3(config-ext-nacl)#deny tcp 192.168.6.0 0.0.0.255 any eq 22
Floor-RTR-3(config-ext-nacl)#deny tcp 192.168.2.0 0.0.0.255 any eq 22
Floor-RTR-3(config-ext-nacl)#permit tcp any any eq 22
Floor-RTR-3(config-ext-nacl)#permit ip any any
Floor-RTR-3(config-ext-nacl)#
Floor-RTR-3(config-ext-nacl)#
Floor-RTR-3(config-ext-nacl)#exit
Floor-RTR-3(config)
```

Assigning on the Interface:

```
Floor-RTR-3(config)#int gi0/4
Floor-RTR-3(config-if)#ip access-group SSH-BLOCK-NON
Floor-RTR-3(config-if)#
Floor-RTR-3(config-if)#ip access-group SSH-BLOCK-NONIT in
Floor-RTR-3(config-if)#
Floor-RTR-3(config-if)#exit
```

Verification Screenshots:


MobaLinux
192.168.221.10:32789
192.168.221.10:32789
VPCS>
VPCS> ip dhcp
DORA IP 192.168.2.2/24 GW 192.168.2.1
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.2.2/24
GATEWAY : 192.168.2.1
DNS : 8.8.8.8
DHCP SERVER : 192.168.9.2
DHCP LEASE : 86395, 86400/43200/75600
DOMAIN NAME : vic.com
MAC : 00:50:79:66:68:15
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS> ping 1.0.0.1
84 bytes from 1.0.0.1 icmp_seq=1 ttl=125 time=39.476 ms
84 bytes from 1.0.0.1 icmp_seq=2 ttl=125 time=143.916 ms
84 bytes from 1.0.0.1 icmp_seq=3 ttl=125 time=64.160 ms
84 bytes from 1.0.0.1 icmp_seq=4 ttl=125 time=31.711 ms
84 bytes from 1.0.0.1 icmp_seq=5 ttl=125 time=123.863 ms
VPCS> tracer 1.0.0.1
trace to 1.0.0.1, 8 hops max, press Ctrl+C to stop
1 192.168.2.1 8.999 ms 6.230 ms 4.096 ms
2 192.168.9.6 8.606 ms 8.217 ms 6.784 ms
3 172.16.88.10 9.428 ms 9.385 ms 8.189 ms
4 192.168.221.2 9.100 ms 10.357 ms 8.142 ms


Oct 21 12:55
71%
Capturing from wlo1
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Version Info
92 14.450632755 192.168.1.45 142.251.155.119 UDP 75 57977 → 443 Len=33
93 14.455269595 142.251.155.119 192.168.1.45 UDP 68 443 → 57977 Len=26
94 14.463923227 142.251.155.119 192.168.1.45 UDP 69 443 → 57977 Len=27
95 15.768123794 192.168.1.44 224.0.0.251 MDNS 119 Standard query 0x0289 PTR _googlecast._tcp.local, "QM" question PTR _674A0243._sub._googlecast._tcp.
96 17.192052580 142.250.66.14 192.168.1.45 TLSv1.2 154 TLS 1.2 Application Data
97 17.192052989 192.168.1.1 224.0.0.1 IGMPv2 46 Membership Query, general
98 17.192368002 192.168.1.45 142.250.66.14 TLSv1.2 101 TLS 1.2 Application Data
99 17.196105189 192.168.1.45 142.250.66.14 TLSv1.2 101 TLS 1.2 Application Data
100 17.232983419 192.168.1.45 142.250.66.14 TCP 101 [TCP Retransmission] 36826 → 443 [PSH, ACK] Seq=36 Ack=89 Win=80 Len=35 TSval=2286208365 TSecr=31979
101 17.237968289 142.250.66.14 192.168.1.45 TCP 66 443 → 36826 [ACK] Seq=89 Ack=36 Win=723 Len=0 TSval=3197958919 TSecr=2286208324
102 17.237968788 142.250.66.14 192.168.1.45 TCP 66 443 → 36826 [ACK] Seq=89 Ack=71 Win=723 Len=0 TSval=3197958919 TSecr=2286208328
103 17.246349966 142.250.66.14 192.168.1.45 TCP 78 [TCP Dup ACK 102#1] 443 → 36826 [ACK] Seq=89 Ack=71 Win=723 Len=0 TSval=3197958955 TSecr=2286208365
104 17.431147929 192.168.1.45 1.0.0.1 ICMP 98 Echo (ping) request id=0x0400, seq=1/256, ttl=60 (reply in 105)
105 17.450294406 1.0.0.1 192.168.1.45 ICMP 98 Echo (ping) reply id=0x0400, seq=1/256, ttl=56 (request in 104)
106 18.460611334 192.168.1.45 1.0.0.1 ICMP 98 Echo (ping) request id=0x0401, seq=2/512, ttl=60 (reply in 107)
107 18.477675121 1.0.0.1 192.168.1.45 ICMP 98 Echo (ping) reply id=0x0401, seq=2/512, ttl=56 (request in 106)
108 19.486562081 192.168.1.45 1.0.0.1 ICMP 98 Echo (ping) request id=0x0402, seq=3/768, ttl=60 (reply in 109)
109 19.506419463 1.0.0.1 192.168.1.45 ICMP 98 Echo (ping) reply id=0x0402, seq=3/768, ttl=56 (request in 108)
110 20.095449486 192.168.1.45 8.8.4.4 DNS 82 Standard query 0x772b HTTPS b1.nel.goog OPT
111 20.095596576 192.168.1.45 8.8.4.4 DNS 82 Standard query 0x9564 A b1.nel.goog OPT
112 20.109286332 8.8.4.4 192.168.1.45 DNS 98 Standard query response 0x9564 A b1.nel.goog A 142.250.205.195 OPT
113 20.115920942 192.168.1.45 142.250.205.195 QUIC 1292 Initial, DCID=71125ad455500619, PKN: 1, PADDING, PING, PADDING, PING, PING, CRYPTO, PING, CRYPTO, CR
114 20.115965071 192.168.1.45 142.250.205.195 QUIC 1292 Initial, DCID=71125ad455500619, PKN: 2, CRYPTO
115 20.115997375 192.168.1.45 142.250.205.195 QUIC 1292 Initial, DCID=71125ad455500619, PKN: 3, PING, CRYPTO, PING, CRYPTO, PADDING, CRYPTO, PADDING, CRYPTO
Frame 1: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface wlo1, id 0
Ethernet II, Src: Digisol_05:c5:10 (54:a2:45:05:c5:10), Dst: AzureWaveTec_ee:f5:67 (f8:54:f6:ee:f5:67)
Internet Protocol Version 4, Src: 148.113.9.222, Dst: 192.168.1.45
Transmission Control Protocol, Src Port: 443, Dst Port: 40450, Seq: 1, Ack: 1, Len: 0
0000 f8 54 f6 ee f5 67 54 a2 45 05 c5 10 08 00 45 00
0010 00 34 ed 74 40 00 33 06 fa 2a 94 71 09 de c0 a8
0020 01 2d 01 bb 9e 02 2d 6c 91 be bf 74 27 5c 80 10
0030 01 f5 5f 2b 00 00 01 01 08 0a d3 43 56 09 cd 5a
0040 79 17
wlo1: <live capture in progress>
Packets: 154
Profile: Default

My learning:

1 Default information originate always we need to give then only in ospf it will work

2 In the pat concept my issue is not choosing a network in the real world. Maybe it will work but here it is the issue. I have enabled the no cef that means cisco express forwarding.

3 is that ACL Seq mismatch so the ssh worked so i changed the sequence and one more thing the where traffic will generate it will not work so initiated the next hop or why i did not implemented in the wireless router means so in the eve-ng i dont have wireless radios so i don thave any lan devices so i changed to floor routers

the acl configurations and seq and inbound and outbound i learned extra